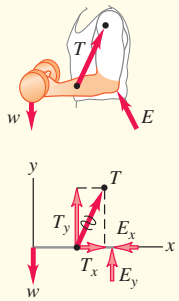
CHAPTER 11 SUMMARY

Conditions for equilibrium: For a rigid body to be in equilibrium, two conditions must be satisfied. First, the vector sum of forces must be zero. Second, the sum of torques about any point must be zero. The torque due to the weight of an object can be found by assuming the entire weight is concentrated at the center of gravity, which is at the same point as the center of mass if $\vec{g}$ has the same value at all points. (See Examples 11.1–11.4.)

$$\sum \vec{F} = \mathbf{0} \qquad (11.1)$$

$$\sum \vec{\tau} = \mathbf{0} \quad \text{about } \textit{any} \text{ point} \qquad (11.2)$$

$$\vec{r}_{\text{cm}} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3 + \cdots}{m_1 + m_2 + m_3 + \cdots} \qquad (11.4)$$

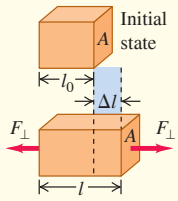


Stress, strain, and Hooke's law: Hooke's law states that in elastic deformations, stress (force per unit area) is proportional to strain (fractional deformation). The proportionality constant is called the elastic modulus.

$$\frac{\text{Stress}}{\text{Strain}} = \text{Elastic modulus} \qquad (11.7)$$

Tensile and compressive stress: Tensile stress is tensile force per unit area, $F_\perp/A$. Tensile strain is fractional change in length, $\Delta l/l_0$. The elastic modulus for tension is called Young's modulus Y. Compressive stress and strain are defined in the same way. (See Example 11.5.)

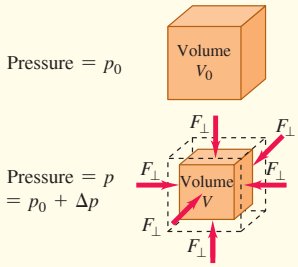
$$Y = \frac{\text{Tensile stress}}{\text{Tensile strain}} = \frac{F_\perp/A}{\Delta l/l_0} = \frac{F_\perp}{A}\frac{l_0}{\Delta l} \qquad (11.10)$$



Bulk stress: Pressure in a fluid is force per unit area. Bulk stress is pressure change, Δp, and bulk strain is fractional volume change, $\Delta V/V_0$. The elastic modulus for compression is called the bulk modulus, B. Compressibility, k, is the reciprocal of bulk modulus: $k = 1/B$. (See Example 11.6.)

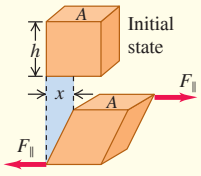
$$p = \frac{F_\perp}{A} \qquad (11.11)$$

$$B = \frac{\text{Bulk stress}}{\text{Bulk strain}} = -\frac{\Delta p}{\Delta V/V_0} \qquad (11.13)$$



Shear stress: Shear stress is force per unit area, $F_\parallel/A$, for a force applied tangent to a surface. Shear strain is the displacement x of one side divided by the transverse dimension h. The elastic modulus for shear is called the shear modulus, S. (See Example 11.7.)

$$S = \frac{\text{Shear stress}}{\text{Shear strain}} = \frac{F_\parallel/A}{x/h} = \frac{F_\parallel}{A}\frac{h}{x} \qquad (11.17)$$



The limits of Hooke's law: The proportional limit is the maximum stress for which stress and strain are proportional. Beyond the proportional limit, Hooke's law is not valid. The elastic limit is the stress beyond which irreversible deformation occurs. The breaking stress, or ultimate strength, is the stress at which the material breaks.

Chapter 11 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLE 11.1 (Section 11.2) before attempting these problems.

VP11.1.1 A uniform plank 8.00 m in length with mass 40.0 kg is supported at two points located 1.00 m and 5.00 m, respectively, from the left-hand end. What is the maximum additional mass you could place on the right-hand end of the plank and have the plank still be at rest?

VP11.1.2 A bowling ball (which we can regard as a uniform sphere) has a mass of 7.26 kg and a radius of 0.216 m. A baseball has a mass of 0.145 kg. If you connect these two balls with a lightweight rod, what must be the distance between the center of the bowling ball and the center of the baseball so that the system of the two balls and the rod will balance at the point where the rod touches the surface of the bowling ball?

VP11.1.3 Three small objects are arranged along a uniform rod of mass m and length L: one of mass m at the left end, one of mass m at the center, and one of mass $2m$ at the right end. How far to the left or right of the rod's center should you place a support so that the rod with the attached objects will balance there?

VP11.1.4 A small airplane with full fuel tanks, but no occupants or baggage, has a mass of 1.17×10^3 kg and a center of gravity that is 2.58 m behind the nose of the airplane. The pilot's seat is 2.67 m behind the nose, and the baggage compartment is 4.30 m behind the nose. A 75.0 kg pilot boards the plane and is the only occupant. If the center of gravity of the airplane with pilot can be no more than 2.76 m behind the nose for in-flight stability, what is the maximum mass that the baggage compartment can hold?

Be sure to review EXAMPLES 11.2, 11.3, and 11.4 (Section 11.3) before attempting these problems.

VP11.4.1 The rear wheels of a truck support 57.0% of the weight of the truck, while the front wheels support 43.0% of the weight. The center of gravity of the truck is 2.52 m in front of the rear wheels. What is the wheelbase of the truck (the distance between the front and rear wheels)?

VP11.4.2 A small airplane is sitting at rest on the ground. Its center of gravity is 2.58 m behind the nose of the airplane, the front wheel (nose wheel) is 0.800 m behind the nose, and the main wheels are 3.02 m behind the nose. What percentage of the airplane's weight is supported by the nose wheel, and what percentage is supported by the main wheels?

VP11.4.3 Figure 11.11 shows a metal advertising sign of weight w suspended from the end of a horizontal rod of negligible mass and length L. The end of the rod with the sign is supported by a cable at an angle θ from the horizontal, and the other end is supported by a hinge at point P. (a) Using the idea that there is zero net torque about the end of the rod with the attached sign, find the vertical component of the force $\vec{F}_{\text{hinge}}$ exerted by the hinge. (b) Using the idea that there is zero net vertical force on the rod with the attached sign, find the tension in the cable. (c) Using the idea that there is zero net horizontal force on the rod with the attached sign, find the horizontal component of the force exerted by the hinge.

VP11.4.4 Suppose that in Figure 11.11 the horizontal rod has weight w (the same as the hanging sign). (a) Using the idea that there is zero net torque about the end of the rod with the attached sign, find the vertical component of the force $\vec{F}_{\text{hinge}}$ exerted by the hinge. (b) Using the idea that there is zero net vertical force on the rod with the attached sign, find the tension in the cable. (c) Using the idea that there is zero net horizontal force on the rod with the attached sign, find the horizontal component of the force exerted by the hinge.

Be sure to review EXAMPLES 11.5, 11.6, and 11.7 (Section 11.4) before attempting these problems.

VP11.7.1 A copper wire has a radius of 4.5 mm. When forces of a certain equal magnitude but opposite directions are applied to the ends of the wire, the wire stretches by 5.0×10^{-3} of its original length. (a) What is the tensile stress on the wire? (b) What is the magnitude of the force on either end?

VP11.7.2 An aluminum cylinder with a radius of 2.5 cm and a height of 82 cm is used as one leg of a workbench. The workbench pushes down on the cylinder with a force of 3.2×10^4 N. (a) What is the compressive strain of the cylinder? (b) By what distance does the cylinder's height decrease as a result of the forces on it?

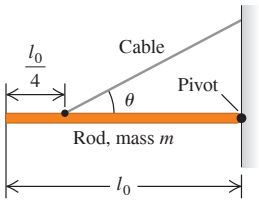
VP11.7.3 The pressure on the surface of a sphere of radius 1.2 cm is increased by 2.5×10^7 Pa. Calculate the resulting decrease in volume of the sphere if it is made (a) of lead and (b) of mercury.

VP11.7.4 You apply forces of magnitude 4.2×10^4 N to the top and bottom surfaces of a brass cube. The forces are tangent to each surface and parallel to the sides of each surface. If the cube is 2.5 cm on a side, what is the resulting shear displacement?

BRIDGING PROBLEM In Equilibrium and Under Stress

A horizontal, uniform, solid copper rod has an original length l_0, cross-sectional area A, Young's modulus Y, bulk modulus B, shear modulus S, and mass m. It is supported by a frictionless pivot at its right end and by a cable a distance $l_0/4$ from its left end (**Fig. 11.21**). Both pivot and cable are attached so that they exert their forces uniformly over the rod's cross section. The cable makes an angle θ with the rod and compresses it. (a) Find the tension in the cable. (b) Find the magnitude and direction of the force exerted by the pivot on the right end of the rod. How does this magnitude compare to the cable tension? How does this angle compare to θ? (c) Find the change in length of the rod due to the stresses exerted by the cable and pivot on the rod. (The length change is small compared to the original length l_0.) (d) By what factor would your answer in part (c) increase if the solid copper rod were twice as long but had the same cross-sectional area?

Figure 11.21 What are the forces on the rod? What are the stress and strain?



SOLUTION GUIDE

IDENTIFY and SET UP

1. Draw a free-body diagram for the rod. Be careful to place each force in the correct location.
2. List the unknown quantities, and decide which are the target variables.
3. What conditions must be met so that the rod remains at rest? What kind of stress (and resulting strain) is involved? Use your answers to select the appropriate equations.

EXECUTE

4. Use your equations to solve for the target variables. (*Hint:* You can make the solution easier by carefully choosing the point around which you calculate torques.)
5. Use trigonometry to decide whether the pivot force or the cable tension has the greater magnitude and whether the angle of the pivot force is greater than, less than, or equal to θ.

EVALUATE

6. Check whether your answers are reasonable. Which force, the cable tension or the pivot force, holds up more of the weight of the rod? Does this make sense?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q11.1 Does a rigid object in uniform rotation about a fixed axis satisfy the first and second conditions for equilibrium? Why? Does it then follow that every particle in this object is in equilibrium? Explain.

Q11.2 (a) Is it possible for an object to be in translational equilibrium (the first condition) but *not* in rotational equilibrium (the second condition)? Illustrate your answer with a simple example. (b) Can an object be in rotational equilibrium yet *not* in translational equilibrium? Justify your answer with a simple example.

Q11.3 Car tires are sometimes "balanced" on a machine that pivots the tire and wheel about the center. Weights are placed around the wheel rim until it does not tip from the horizontal plane. Discuss this procedure in terms of the center of gravity.

Q11.4 Does the center of gravity of a solid object always lie within the material of the object? If not, give a counterexample.

Q11.5 In Section 11.2 we always assumed that the value of g was the same at all points on the object. This is *not* a good approximation if the dimensions of the object are great enough, because the value of g decreases with altitude. If this is taken into account, will the center of gravity of a long, vertical rod be above, below, or at its center of mass? Explain how this can be used to keep the long axis of an orbiting spacecraft pointed toward the earth. (This would be useful for a weather satellite that must always keep its camera lens trained on the earth.) The moon is not exactly spherical but is somewhat elongated. Explain why this same effect is responsible for keeping the same face of the moon pointed toward the earth at all times.

Q11.6 You are balancing a wrench by suspending it at a single point. Is the equilibrium stable, unstable, or neutral if the point is above, at, or below the wrench's center of gravity? In each case give the reasoning behind your answer. (For rotation, a rigid object is in *stable* equilibrium if a small rotation of the object produces a torque that tends to return the object to equilibrium; it is in *unstable* equilibrium if a small rotation produces a torque that tends to take the object farther from equilibrium; and it is in *neutral* equilibrium if a small rotation produces no torque.)

Q11.7 You can probably stand flatfooted on the floor and then rise up and balance on your tiptoes. Why are you unable do it if your toes are touching the wall of your room? (Try it!)

Q11.8 You freely pivot a horseshoe from a horizontal nail through one of its nail holes. You then hang a long string with a weight at its bottom from the same nail, so that the string hangs vertically in front of the horseshoe without touching it. How do you know that the horseshoe's center of gravity is along the line behind the string? How can you locate the center of gravity by repeating the process at another nail hole? Will the center of gravity be within the solid material of the horseshoe?

Q11.9 An object consists of a ball of weight W glued to the end of a uniform bar also of weight W. If you release it from rest, with the bar horizontal, what will its behavior be as it falls if air resistance is negligible? Will it (a) remain horizontal; (b) rotate about its center of gravity; (c) rotate about the ball; or (d) rotate so that the ball swings downward? Explain your reasoning.

Q11.10 Suppose that the object in Question 11.9 is released from rest with the bar tilted at 60° above the horizontal with the ball at the upper end. As it is falling, will it (a) rotate about its center of gravity until it is horizontal; (b) rotate about its center of gravity until it is vertical with the ball at the bottom; (c) rotate about the ball until it is vertical with the ball at the bottom; or (d) remain at 60° above the horizontal?

Q11.11 Why must a water skier moving with constant velocity lean backward? What determines how far back she must lean? Draw a free-body diagram for the water skier to justify your answers.

Q11.12 In pioneer days, when a Conestoga wagon was stuck in the mud, people would grasp the wheel spokes and try to turn the wheels, rather than simply pushing the wagon. Why?

Q11.13 The mighty Zimbo claims to have leg muscles so strong that he can stand flat on his feet and lean forward to pick up an apple on the floor with his teeth. Should you pay to see him perform, or do you have any suspicions about his claim? Why?

Q11.14 Why is it easier to hold a 10 kg dumbbell in your hand at your side than it is to hold it with your arm extended horizontally?

Q11.15 Certain features of a person, such as height and mass, are fixed (at least over relatively long periods of time). Are the following features also fixed? (a) location of the center of gravity of the body; (b) moment of inertia of the body about an axis through the person's center of mass. Explain your reasoning.

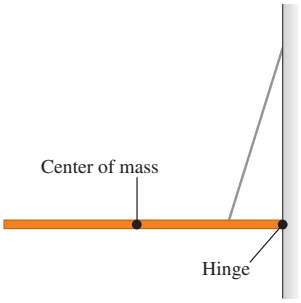
Q11.16 During pregnancy, women often develop back pains from leaning backward while walking. Why do they have to walk this way?

Q11.17 Why is a tapered water glass with a narrow base easier to tip over than a glass with straight sides? Does it matter whether the glass is full or empty?

Q11.18 When a tall, heavy refrigerator is pushed across a rough floor, what factors determine whether it slides or tips?

Q11.19 A uniform beam is suspended horizontally and attached to a wall by a small hinge (**Fig. Q11.19**). What are the directions (upward or downward, and to the left or the right) of the components of the force that the hinge exerts *on the beam*? Explain.

Figure **Q11.19**



Q11.20 If a metal wire has its length doubled and its diameter tripled, by what factor does its Young's modulus change?

Q11.21 A metal wire of diameter D stretches by 0.100 mm when supporting a weight W. If the same-length wire is used to support a weight three times as heavy, what would its diameter have to be (in terms of D) so it still stretches only 0.100 mm?

Q11.22 Compare the mechanical properties of a steel cable, made by twisting many thin wires together, with the properties of a solid steel rod of the same diameter. What advantages does each have?

Q11.23 The material in human bones and elephant bones is essentially the same, but an elephant has much thicker legs. Explain why, in terms of breaking stress.

Q11.24 There is a small but appreciable amount of elastic hysteresis in the large tendon at the back of a horse's leg. Explain how this can cause damage to the tendon if a horse runs too hard for too long a time.

Q11.25 When rubber mounting blocks are used to absorb machine vibrations through elastic hysteresis, as mentioned in Section 11.5, what becomes of the energy associated with the vibrations?

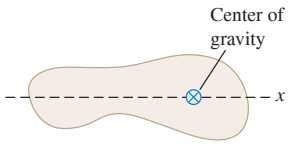
EXERCISES

Section 11.2 Center of Gravity

11.1 •• A 0.140 kg, 42.1-cm-long uniform bar has a small 0.055 kg mass glued to its left end and a small 0.150 kg mass glued to the other end. The two small masses can each be treated as point masses. You want to balance this system horizontally on a fulcrum placed just under its center of gravity. How far from the left end should the fulcrum be placed?

11.2 •• The center of gravity of a 4.00 kg irregular object is shown in **Fig. E11.2.** You need to move the center of gravity 3.00 cm to the left by gluing on a 1.70 kg mass, which will then be considered as part of the object. Where should the center of gravity of this additional mass be located?

Figure **E11.2**



11.3 • A uniform rod is 2.20 m long and has mass 1.50 kg. A 2.20 kg clamp is attached to the rod. How far should the center of gravity of the clamp be from the left-hand end of the rod in order for the center of gravity of the composite object to be 1.30 m from the left-hand end of the rod?

11.4 • Consider the free-body diagram shown in Fig. 11.9b. (a) What is the horizontal distance of the center of gravity of the person–ladder system from the point where the ladder touches the ground? (b) What is the torque about the rotation axis shown in the figure (point B) computed by taking the total weight of the person plus ladder acting at the center of gravity? (c) How does the result in part (b) compare to the sum of the individual torques computed in Example 11.3 for the weight of the person and the weight of the ladder?

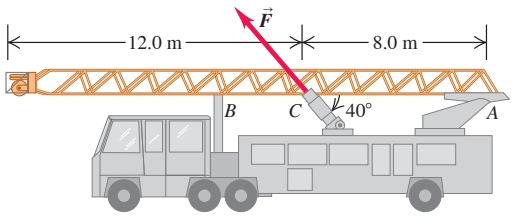
11.5 •• A uniform steel rod has mass 0.300 kg and length 40.0 cm and is horizontal. A uniform sphere with radius 8.00 cm and mass 0.900 kg is welded to one end of the bar, and a uniform sphere with radius 6.00 cm and mass 0.380 kg is welded to the other end of the bar. The centers of the rod and of each sphere all lie along a horizontal line. How far is the center of gravity of the combined object from the center of the rod?

Section 11.3 Solving Rigid-Body Equilibrium Problems

11.6 • A uniform 300 N trapdoor in a floor is hinged at one side. Find the net upward force needed to begin to open it and the total force exerted on the door by the hinges (a) if the upward force is applied at the center and (b) if the upward force is applied at the center of the edge opposite the hinges.

11.7 •• **Raising a Ladder.** A ladder carried by a fire truck is 20.0 m long. The ladder weighs 3100 N and its center of gravity is at its center. The ladder is pivoted at one end (A) about a pin (**Fig. E11.7**); ignore the friction torque at the pin. The ladder is raised into position by a force applied by a hydraulic piston at C. Point C is 8.0 m from A, and the force $\vec{F}$ exerted by the piston makes an angle of 40° with the ladder. What magnitude must $\vec{F}$ have to just lift the ladder off the support bracket at B? Start with a free-body diagram of the ladder.

Figure **E11.7**

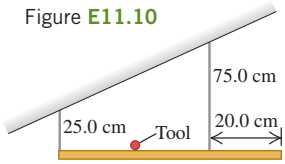


11.8 •• Two people are carrying a uniform wooden board that is 3.00 m long and weighs 160 N. If one person applies an upward force equal to 60 N at one end, at what point does the other person lift? Begin with a free-body diagram of the board.

11.9 •• Two people carry a heavy electric motor by placing it on a light board 2.70 m long. One person lifts at one end with a force of 370 N, and the other lifts the opposite end with a force of 700 N. (a) What is the weight of the motor, and where along the board is its center of gravity located? (b) Suppose the board is not light but weighs 210 N, with its center of gravity at its center, and the two people each exert the same forces as before. What is the weight of the motor in this case, and where is its center of gravity located?

11.10 •• An 80.0 cm, uniform, 59.0 N shelf is supported horizontally by two vertical wires attached to the sloping ceiling (**Fig. E11.10**). A very small 18.0 N tool is placed on the shelf midway between the points where the wires are attached to it. Find the tension in each wire. Begin by making a free-body diagram of the shelf.

Figure **E11.10**

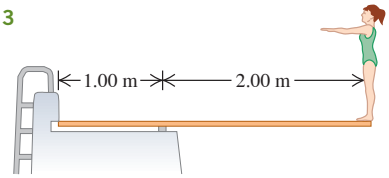


11.11 •• A 362 N, uniform, 1.99 m bar is suspended horizontally by two vertical cables at each end. Cable *A* can support a maximum tension of 511.0 N without breaking, and cable *B* can support up to 356.0 N. You want to place a small weight on this bar. (a) What is the heaviest weight you can put on without breaking either cable, and (b) where should you put this weight?

11.12 •• A uniform ladder 5.0 m long rests against a frictionless, vertical wall with its lower end 3.0 m from the wall. The ladder weighs 160 N. The coefficient of static friction between the foot of the ladder and the ground is 0.40. A man weighing 740 N climbs slowly up the ladder. Start by drawing a free-body diagram of the ladder. (a) What is the maximum friction force that the ground can exert on the ladder at its lower end? (b) What is the actual friction force when the man has climbed 1.0 m along the ladder? (c) How far along the ladder can the man climb before the ladder starts to slip?

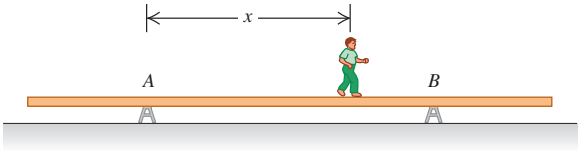
11.13 • A diving board 3.00 m long is supported at a point 1.00 m from the end, and a diver weighing 500 N stands at the free end (**Fig. E11.13**). The diving board is of uniform cross section and weighs 280 N. Find (a) the force at the support point and (b) the force at the left-hand end.

Figure **E11.13**



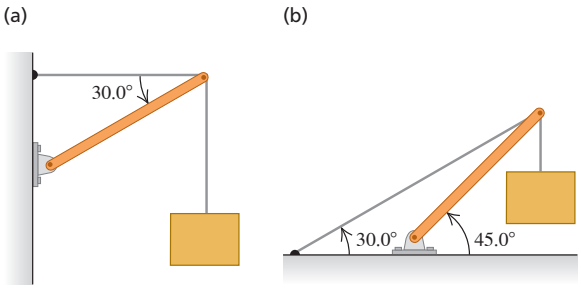
11.14 • A uniform aluminum beam 9.00 m long, weighing 300 N, rests symmetrically on two supports 5.00 m apart (**Fig. E11.14**). A boy weighing 600 N starts at point *A* and walks toward the right. (a) In the same diagram construct two graphs showing the upward forces F_A and F_B exerted on the beam at points *A* and *B*, as functions of the coordinate *x* of the boy. Let 1 cm = 100 N vertically and 1 cm = 1.00 m horizontally. (b) From your diagram, how far beyond point *B* can the boy walk before the beam tips? (c) How far from the right end of the beam should support *B* be placed so that the boy can walk just to the end of the beam without causing it to tip?

Figure **E11.14**



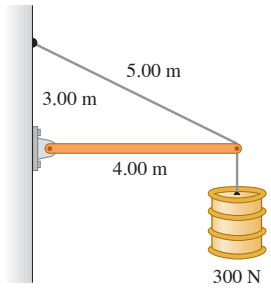
11.15 • Find the tension *T* in each cable and the magnitude and direction of the force exerted on the strut by the pivot in each of the arrangements in **Fig. E11.15**. In each case let *w* be the weight of the suspended crate full of priceless art objects. The strut is uniform and also has weight *w*. Start each case with a free-body diagram of the strut.

Figure **E11.15**



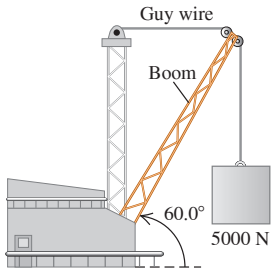
11.16 • The horizontal beam in **Fig. E11.16** weighs 190 N, and its center of gravity is at its center. Find (a) the tension in the cable and (b) the horizontal and vertical components of the force exerted on the beam at the wall.

Figure **E11.16**



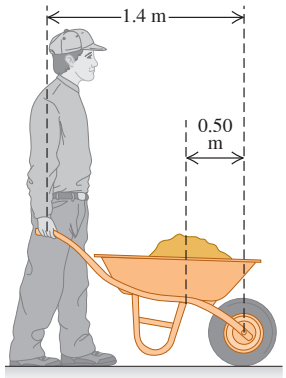
11.17 •• The boom shown in **Fig. E11.17** weighs 2600 N and is attached to a frictionless pivot at its lower end. It is not uniform; the distance of its center of gravity from the pivot is 35% of its length. Find (a) the tension in the guy wire and (b) the horizontal and vertical components of the force exerted on the boom at its lower end. Start with a free-body diagram of the boom.

Figure **E11.17**



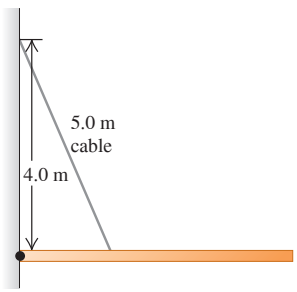
11.18 •• You are pushing an 80.0 N wheelbarrow as shown in **Fig. E11.18**. You lift upward on the handle of the wheelbarrow so that the only point of contact between the wheelbarrow and the ground is at the front wheel. Assume the distances are as shown in the figure, where 0.50 m is the horizontal distance from the center of the wheel to the center of gravity of the wheelbarrow. The center of gravity of the dirt in the wheelbarrow is assumed to also be a horizontal distance of 0.50 m from the center of the wheel. Estimate the maximum total upward force that you can apply to the wheelbarrow handles. (a) If you apply this estimated force, what is the maximum weight of dirt that you can carry in the wheelbarrow? (b) If the dirt has the weight you calculated in part (a), what upward force does the ground apply to the wheel of the wheelbarrow?

Figure **E11.18**



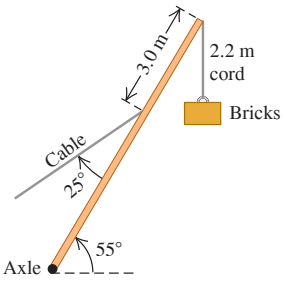
11.19 •• A 9.00-m-long uniform beam is hinged to a vertical wall and held horizontally by a 5.00-m-long cable attached to the wall 4.00 m above the hinge (**Fig. E11.19**). The metal of this cable has a test strength of 1.00 kN, which means that it will break if the tension in it exceeds that amount. (a) Draw a free-body diagram of the beam. (b) What is the heaviest beam that the cable can support in this configuration? (c) Find the horizontal and vertical components of the force the hinge exerts on the beam. Is the vertical component upward or downward?

Figure **E11.19**



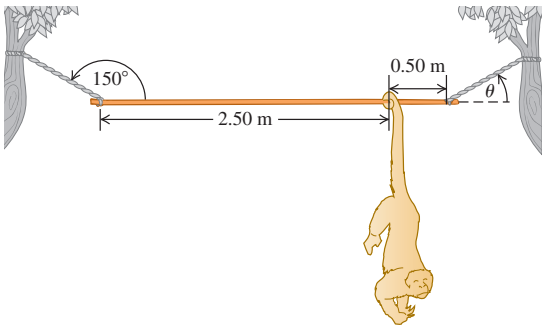
11.20 •• A 15,000 N crane pivots around a friction-free axle at its base and is supported by a cable making a 25° angle with the crane (**Fig. E11.20**). The crane is 16 m long and is not uniform, its center of gravity being 7.0 m from the axle as measured along the crane. The cable is attached 3.0 m from the upper end of the crane. When the crane is raised to 55° above the horizontal holding an 11,000 N pallet of bricks by a 2.2 m, very light cord, find (a) the tension in the cable and (b) the horizontal and vertical components of the force that the axle exerts on the crane. Start with a free-body diagram of the crane.

Figure E11.20



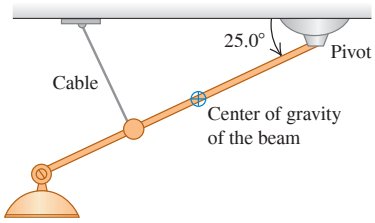
11.21 •• A 3.00-m-long, 290 N, uniform rod at the zoo is held in a horizontal position by two ropes at its ends (**Fig. E11.21**). The left rope makes an angle of 150° with the rod, and the right rope makes an angle θ with the horizontal. A 86 N howler monkey (*Alouatta seniculus*) hangs motionless 0.50 m from the right end of the rod as he carefully studies you. Calculate the tensions in the two ropes and the angle θ. First make a free-body diagram of the rod.

Figure E11.21



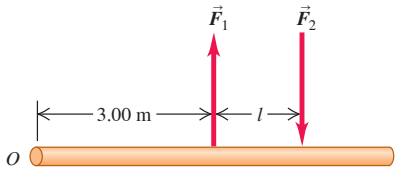
11.22 •• A nonuniform beam 4.45 m long and weighing 1.10 kN makes an angle of 25.0° below the horizontal. It is held in position by a frictionless pivot at its upper right end and by a cable 3.10 m farther down the beam and perpendicular to it (**Fig. E11.22**). The center of gravity of the beam is 2.10 m down the beam from the pivot. Lighting equipment exerts a 4.93 kN downward force on the lower left end of the beam. Find the tension T in the cable and the horizontal and vertical components of the force exerted on the beam by the pivot. Start by sketching a free-body diagram of the beam.

Figure E11.22



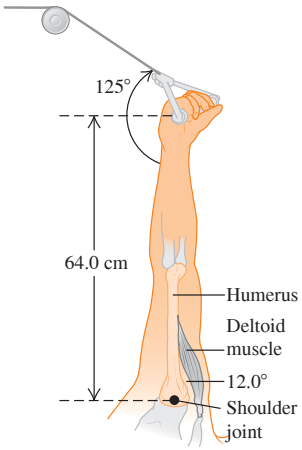
11.23 • **A Couple.** Two forces equal in magnitude and opposite in direction, acting on an object at two different points, form what is called a *couple*. Two antiparallel forces with equal magnitudes $F_1 = F_2 = 6.30$ N are applied to a rod as shown in **Fig. E11.23**. (a) What should the distance l between the forces be if they are to provide a net torque of 6.50 N · m about the left end of the rod? (b) Is the sense of this torque clockwise or counterclockwise? (c) Repeat parts (a) and (b) for a pivot at the point on the rod where $\vec{F}_2$ is applied.

Figure E11.23



11.24 •• BIO **A Good Workout.** You are doing exercises on a Nautilus machine in a gym to strengthen your deltoid (shoulder) muscles. Your arms are raised vertically and can pivot around the shoulder joint, and you grasp the cable of the machine in your hand 64.0 cm from your shoulder joint. The deltoid muscle is attached to the humerus 15.0 cm from the shoulder joint and makes a 12.0° angle with that bone (**Fig. E11.24**). If you have set the tension in the cable of the machine to 36.0 N on each arm, what is the tension in each deltoid muscle if you simply hold your outstretched arms in place? (*Hint:* Start by making a clear free-body diagram of your arm.)

Figure E11.24



11.25 •• A uniform rod has one end attached to a vertical wall by a frictionless hinge. A horizontal wire runs from the other end of the rod to a point on the wall above the hinge and holds the rod at an angle θ above the horizontal. You vary the angle θ by changing the length of the wire, and for each θ you measure the tension T in the wire. You plot your data in the form of a T-versus-$\cot\theta$ graph. The data lie close to a straight line that has slope 30.0 N. What is the mass of the rod?

Section 11.4 Stress, Strain, and Elastic Moduli

11.26 • BIO **Biceps Muscle.** A relaxed biceps muscle requires a force of 25.0 N for an elongation of 3.0 cm; the same muscle under maximum tension requires a force of 500 N for the same elongation. Find Young's modulus for the muscle tissue under each of these conditions if the muscle is assumed to be a uniform cylinder with length 0.200 m and cross-sectional area 50.0 cm^2.

11.27 •• A circular steel wire 1.98 m long must stretch no more than 0.24 cm when a tensile force of 440 N is applied to each end of the wire. What minimum diameter is required for the wire?

11.28 •• Two cylindrical rods, one steel and the other copper, are joined end to end. Each rod is 0.750 m long and 1.50 cm in diameter. The combination is subjected to a tensile force with magnitude 4000 N. For each rod, what are (a) the strain and (b) the elongation?

11.29 •• A metal rod that is 5.00 m long and 0.54 cm^2 in cross-sectional area is found to stretch 0.11 cm under a tension of 4600 N. What is Young's modulus for this metal?

11.30 •• **Stress on a Mountaineer's Rope.** A nylon rope used by mountaineers elongates 1.10 m under the weight of a 65.0 kg climber. If the rope is 45.0 m in length and 7.0 mm in diameter, what is Young's modulus for nylon?

11.31 • A lead sphere has volume 6.0 cm^3 when it is resting on a lab table, where the pressure applied to the sphere is atmospheric pressure. The sphere is then placed in the fluid of a hydraulic press. What increase in the pressure above atmospheric pressure produces a 0.50% decrease in the volume of the sphere?

11.32 •• A vertical, solid steel post 29 cm in diameter and 2.10 m long is required to support a load of 8200 kg. You can ignore the weight of the post. What are (a) the stress in the post; (b) the strain in the post; and (c) the change in the post's length when the load is applied?

11.33 •• BIO **Compression of Human Bone.** The bulk modulus for bone is 15 GPa. (a) If a diver-in-training is put into a pressurized suit, by how much would the pressure have to be raised (in atmospheres) above atmospheric pressure to compress her bones by 0.15% of their original volume? (b) Given that the pressure in the ocean increases by 1.0×10^4 Pa for every meter of depth below the surface, how deep would this diver have to go for her bones to compress by 0.15%? Does it seem that bone compression is a problem she needs to be concerned with when diving?

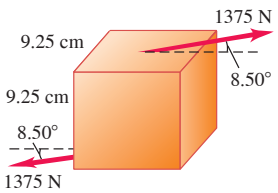
11.34 • A solid gold bar is pulled up from the hold of the sunken RMS *Titanic*. (a) What happens to its volume as it goes from the pressure at the ship to the lower pressure at the ocean's surface? (b) The pressure difference is proportional to the depth. How many times greater would the volume change have been had the ship been twice as deep? (c) The bulk modulus of lead is one-fourth that of gold. Find the ratio of the volume change of a solid lead bar to that of a gold bar of equal volume for the same pressure change.

11.35 • A specimen of oil having an initial volume of 590 cm^3 is subjected to a pressure increase of 4.0×10^6 Pa, and the volume is found to decrease by 0.47 cm^3. What is the bulk modulus of the material? The compressibility?

11.36 •• In the Challenger Deep of the Marianas Trench, the depth of seawater is 10.9 km and the pressure is 1.16×10^8 Pa (about 1.15×10^3 atm). (a) If a cubic meter of water is taken from the surface to this depth, what is the change in its volume? (Normal atmospheric pressure is about 1.0×10^5 Pa. Assume that k for seawater is the same as the freshwater value given in Table 11.2.) (b) What is the density of seawater at this depth? (At the surface, seawater has a density of 1.03×10^3 kg/m^3.)

11.37 •• A square steel plate is 10.0 cm on a side and 0.500 cm thick. (a) Find the shear strain that results if two forces, each of magnitude 9.0×10^5 N and in opposite directions, act tangent to the surfaces of a pair of opposite sides of the object, as in Fig. 11.18. (b) Find the displacement x in centimeters.

Figure E11.38



11.38 • In lab tests on a 9.25 cm cube of a certain material, a force of 1375 N directed at 8.50° to the cube (**Fig. E11.38**) causes the cube to deform through an angle of 1.24°. What is the shear modulus of the material?

11.39 •• A steel wire with radius r_{steel} has a fractional increase in length of $(\Delta l/l_0)_{steel}$ when the tension in the wire is increased from zero to T_{steel}. An aluminum wire has radius r_{al} that is twice the radius of the steel wire: $r_{al} = 2r_{steel}$. In terms of T_{steel}, what tension in the aluminum wire produces the same fractional change in length as in the steel wire?

11.40 •• You apply a force of magnitude $F_\perp$ to one end of a wire and another force $F_\perp$ in the opposite direction to the other end of the wire. The cross-sectional area of the wire is 8.00 mm^2. You measure the fractional change in the length of the wire, $\Delta l/l_0$, for several values of $F_\perp$. When you plot your data with $\Delta l/l_0$ on the vertical axis and $F_\perp$ (in units of N) on the horizontal axis, the data lie close to a line that has slope 8.0×10^{-7} N^{-1}. What is the value of Young's modulus for this wire?

11.41 • An increase in applied pressure Δp_1 produces a fractional volume change of $(\Delta V/V_0)_1$ for a sample of glycerin. In terms of Δp_1, what pressure increase above atmospheric pressure is required to produce the same volume change $(\Delta V/V_0)_1$ for a sample of ethyl alcohol?

Section 11.5 Elasticity and Plasticity

11.42 •• A brass wire is to withstand a tensile force of 350 N without breaking. What minimum diameter must the wire have?

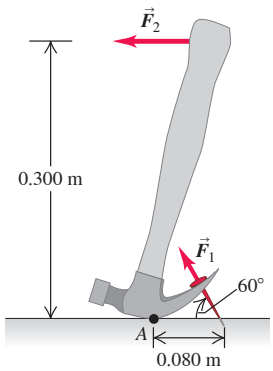
11.43 •• In a materials testing laboratory, a metal wire made from a new alloy is found to break when a tensile force of 87.8 N is applied perpendicular to each end. If the diameter of the wire is 1.19 mm, what is the breaking stress of the alloy?

11.44 •• CP A steel cable with cross-sectional area 2.70 cm^2 has an elastic limit of 2.40×10^8 Pa. Find the maximum upward acceleration that can be given a 1190 kg lift supported by the cable if the stress is not to exceed one-third of the elastic limit.

PROBLEMS

11.45 •• You are using a hammer to pull a nail from the floor, as shown in **Fig. P11.45**. Force $\vec{F}_2$ is the force you apply to the handle, and force $\vec{F}_1$ is the force the hammer applies to the nail. Estimate the maximum magnitude of the force $\vec{F}_2$ that you could apply to the hammer. If you apply this force, what is the magnitude of the force $\vec{F}_1$ that the hammer applies to the nail?

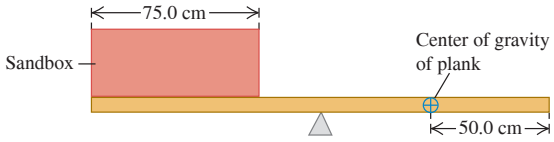
Figure P11.45



11.46 ••• A door 1.05 m wide and 2.20 m high weighs 284 N and is supported by two hinges, one 0.50 m from the top and the other 0.50 m from the bottom. Each hinge supports half the total weight of the door. Assuming that the door's center of gravity is at its center, find the horizontal components of force exerted on the door by each hinge.

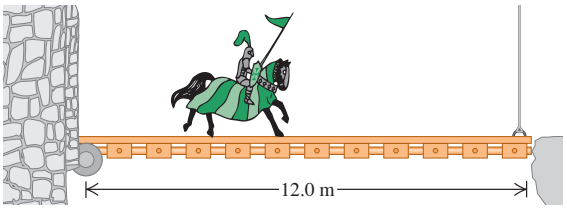
11.47 ••• A box of negligible mass rests at the left end of a 2.00 m, 25.0 kg plank (**Fig. P11.47**). The width of the box is 75.0 cm, and sand is to be distributed uniformly throughout it. The center of gravity of the nonuniform plank is 50.0 cm from the right end. What mass of sand should be put into the box so that the plank balances horizontally on a fulcrum placed just below its midpoint?

Figure P11.47



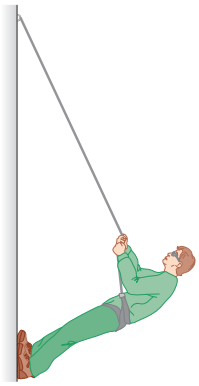
11.48 • Sir Lancelot rides slowly out of the castle at Camelot and onto the 12.0-m-long drawbridge that passes over the moat (**Fig. P11.48**). Unbeknownst to him, his enemies have partially severed the vertical cable holding up the front end of the bridge so that it will break under a tension of 5.80×10^3 N. The bridge has mass 200 kg and its center of gravity is at its center. Lancelot, his lance, his armor, and his horse together have a combined mass of 600 kg. Will the cable break before Lancelot reaches the end of the drawbridge? If so, how far from the castle end of the bridge will the center of gravity of the horse plus rider be when the cable breaks?

Figure **P11.48**



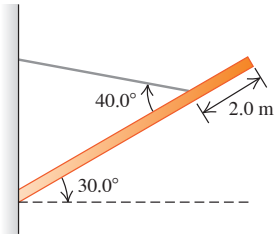
11.49 ••• **Mountain Climbing.** Mountaineers often use a rope to lower themselves down the face of a cliff (this is called *rappelling*). They do this with their body nearly horizontal and their feet pushing against the cliff (**Fig. P11.49**). Suppose that an 82.0 kg climber, who is 1.90 m tall and has a center of gravity 1.1 m from his feet, rappels down a vertical cliff with his body raised 35.0° above the horizontal. He holds the rope 1.40 m from his feet, and it makes a 25.0° angle with the cliff face. (a) What tension does his rope need to support? (b) Find the horizontal and vertical components of the force that the cliff face exerts on the climber's feet. (c) What minimum coefficient of static friction is needed to prevent the climber's feet from slipping on the cliff face if he has one foot at a time against the cliff?

Figure **P11.49**



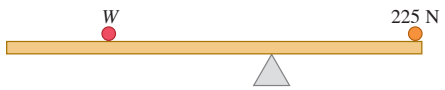
11.50 •• A uniform, 8.0 m, 1150 kg beam is hinged to a wall and supported by a thin cable attached 2.0 m from the free end of the beam (**Fig. P11.50**). The beam is supported at an angle of 30.0° above the horizontal. (a) Draw a free-body diagram of the beam. (b) Find the tension in the cable. (c) How hard does the beam push inward on the wall?

Figure **P11.50**



11.51 •• A uniform, 251 N rod that is 1.80 m long carries a 225 N weight at its right end and an unknown weight W toward the left end (**Fig. P11.51**). When W is placed 47.2 cm from the left end of the rod, the system just balances horizontally when the fulcrum is located 67.3 cm from the right end. (a) Find W. (b) If W is now moved 29.1 cm to the right, how far and in what direction must the fulcrum be moved to restore balance?

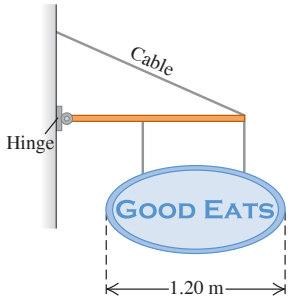
Figure **P11.51**



11.52 ••• A claw hammer is used to pull a nail out of a board (see Fig. P11.45). The nail is at an angle of 60° to the board, and a force $\vec{F}_1$ of magnitude 400 N applied to the nail is required to pull it from the board. The hammer head contacts the board at point A, which is 0.080 m from where the nail enters the board. A horizontal force $\vec{F}_2$ is applied to the hammer handle at a distance of 0.300 m above the board. What magnitude of force $\vec{F}_2$ is required to apply the required 400 N force (F_1) to the nail? (Ignore the weight of the hammer.)

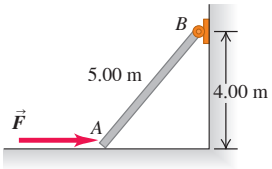
11.53 •• You open a restaurant and hope to entice customers by hanging out a sign (**Fig. P11.53**). The uniform horizontal beam supporting the sign is 1.50 m long, has a mass of 19.0 kg, and is hinged to the wall. The sign itself is uniform with a mass of 35.0 kg and overall length of 1.20 m. The two wires supporting the sign are each 32.0 cm long, are 95.0 cm apart, and are equally spaced from the middle of the sign. The cable supporting the beam is 2.20 m long. (a) What minimum tension must your cable be able to support without having your sign come crashing down? (b) What minimum vertical force must the hinge be able to support without pulling out of the wall?

Figure **P11.53**



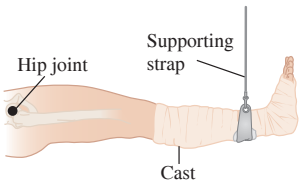
11.54 • End A of the bar AB in **Fig. P11.54** rests on a frictionless horizontal surface, and end B is hinged. A horizontal force $\vec{F}$ of magnitude 102 N is exerted on end A. Ignore the weight of the bar. What are the horizontal and vertical components of the force exerted by the bar on the hinge at B?

Figure **P11.54**



11.55 •• BIO **Supporting a Broken Leg.** A therapist tells a 70 kg patient with a broken leg that he must have his leg in a cast suspended horizontally. For minimum discomfort, the leg should be supported by a vertical strap attached at the center of mass of the leg–cast system (**Fig. P11.55**). To comply with these instructions, the patient consults a table of typical mass distributions and finds that both upper legs (thighs) together typically account for 21.5% of body weight and the center of mass of each thigh is 18.0 cm from the hip joint. The patient also reads that the two lower legs (including the feet) are 14.0% of body weight, with a center of mass 69.0 cm from the hip joint. The cast has a mass of 5.40 kg, and its center of mass is 78.0 cm from the hip joint. How far from the hip joint should the supporting strap be attached to the cast?

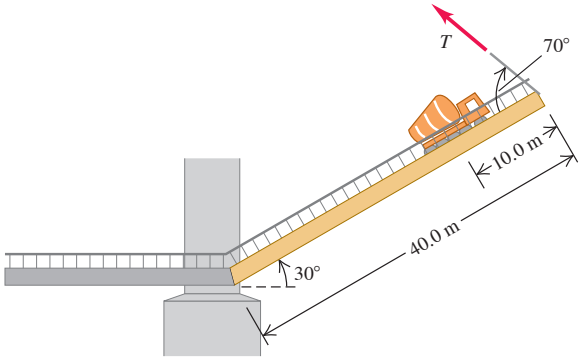
Figure **P11.55**



11.56 • **A Truck on a Drawbridge.** A loaded cement mixer drives onto an old drawbridge, where it stalls with its center of gravity three-quarters of the way across the span. The truck driver radios for help, sets the handbrake, and waits. Meanwhile, a boat approaches, so the drawbridge is raised by means of a cable attached to the end opposite the hinge (**Fig. P11.56**). The drawbridge is 40.0 m long and has a mass of 18,000 kg; its center of gravity is at its midpoint. The cement mixer, with driver, has mass 30,000 kg. When the drawbridge has been raised to an angle of 30° above the horizontal, the cable makes an angle of

70° with the surface of the bridge. (a) What is the tension T in the cable when the drawbridge is held in this position? (b) What are the horizontal and vertical components of the force the hinge exerts on the span?

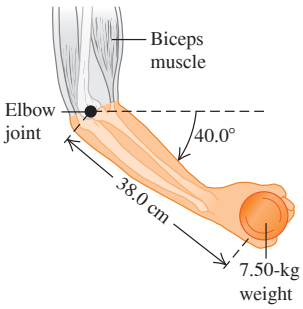
Figure P11.56



11.57 •• CP The left-hand end of a uniform rod of length L and mass m is attached to a vertical wall by a frictionless hinge. The rod is held at an angle θ above the horizontal by a horizontal wire that runs between the wall and the right-hand end of the rod. (a) If the tension in the wire is T, what is the magnitude of the angle θ that the rod makes with the horizontal? (b) The wire breaks and the rod rotates about the hinge. What is the angular speed of the rod as the rod passes through a horizontal position?

11.58 •• BIO **Pumping Iron.** A 72.0 kg weightlifter doing arm raises holds a 7.50 kg weight. Her arm pivots around the elbow joint, starting 40.0° below the horizontal (**Fig. P11.58**). Biometric measurements have shown that, together, the forearms and the hands account for 6.00% of a person's weight. Since the upper arm is held vertically, the biceps muscle always acts vertically and is attached to the bones of the forearm 5.50 cm from the elbow joint. The center of mass of this person's forearm–hand combination is 16.0 cm from the elbow joint, along the bones of the forearm, and she holds the weight 38.0 cm from her elbow joint. (a) Draw a free-body diagram of the forearm. (b) What force does the biceps muscle exert on the forearm? (c) Find the magnitude and direction of the force that the elbow joint exerts on the forearm. (d) As the weightlifter raises her arm toward a horizontal position, will the force in the biceps muscle increase, decrease, or stay the same? Why?

Figure P11.58



11.59 •• The left-hand end of a light rod of length L is attached to a vertical wall by a frictionless hinge. An object of mass m is suspended from the rod at a point a distance αL from the hinge, where $0 < \alpha \leq 1.00$. The rod is held in a horizontal position by a light wire that runs from the right-hand end of the rod to the wall. The wire makes an angle θ with the rod. (a) What is the angle β that the net force exerted by the hinge on the rod makes with the horizontal? (b) What is the value of α for which $\beta = \theta$? (c) What is β when $\alpha = 1.00$?

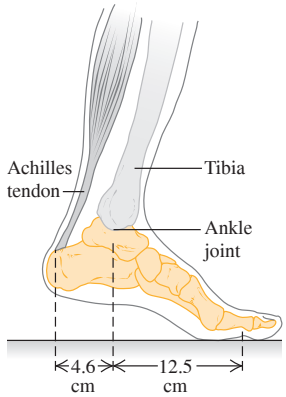
11.60 •• The left-hand end of a slender uniform rod of mass m is placed against a vertical wall. The rod is held in a horizontal position by friction at the wall and by a light wire that runs from the right-hand end of the rod to a point on the wall above the rod. The wire makes an angle θ with the rod. (a) What must the magnitude of the friction force be in order for the rod to remain at rest? (b) If the coefficient of static friction between the rod and the wall is μ_s, what is the maximum angle between the wire and the rod at which the rod doesn't slip at the wall?

11.61 •• A uniform, 6.6-m-long beam weighing 5180 N is hinged to a wall and supported by a thin cable attached 1.8 m from the free end of the beam. The cable runs between the beam and the wall and makes a 47° angle with the beam. What is the tension in the cable when the beam is at an angle of 28° above the horizontal?

11.62 •• CP A uniform drawbridge must be held at a 37° angle above the horizontal to allow ships to pass underneath. The drawbridge weighs 45,000 N and is 14.0 m long. A cable is connected 3.5 m from the hinge where the bridge pivots (measured along the bridge) and pulls horizontally on the bridge to hold it in place. (a) What is the tension in the cable? (b) Find the magnitude and direction of the force the hinge exerts on the bridge. (c) If the cable suddenly breaks, what is the magnitude of the angular acceleration of the drawbridge just after the cable breaks? (d) What is the angular speed of the drawbridge as it becomes horizontal?

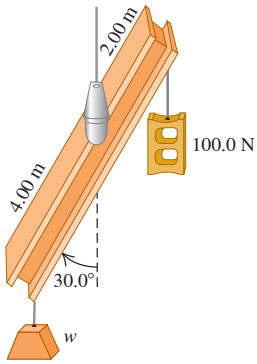
11.63 •• BIO **Tendon-Stretching Exercises.** As part of an exercise program, a 75 kg person does toe raises in which he raises his entire body weight on the ball of one foot (**Fig. P11.63**). The Achilles tendon pulls straight upward on the heel bone of his foot. This tendon is 25 cm long and has a cross-sectional area of 78 mm^2 and a Young's modulus of 1470 MPa. (a) Draw a free-body diagram of the person's foot (everything below the ankle joint). Ignore the weight of the foot. (b) What force does the Achilles tendon exert on the heel during this exercise? Express your answer in newtons and in multiples of his weight. (c) By how many millimeters does the exercise stretch his Achilles tendon?

Figure P11.63



11.64 •• (a) In **Fig. P11.64** a 6.00-m-long, uniform beam is hanging from a point 1.00 m to the right of its center. The beam weighs 140 N and makes an angle of 30.0° with the vertical. At the right-hand end of the beam a 100.0 N weight is hung; an unknown weight w hangs at the left end. If the system is in equilibrium, what is w? You can ignore the thickness of the beam. (b) If the beam makes, instead, an angle of 45.0° with the vertical, what is w?

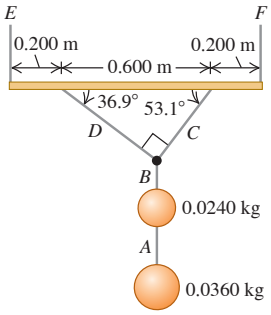
Figure P11.64



11.65 •• The left-hand end of a uniform rod of mass 2.00 kg and length 1.20 m is attached to a vertical wall by a frictionless hinge. The rod is held in a horizontal position by an aluminum wire that runs between the right-hand end of the rod and a point on the wall that is above the hinge. The cross-sectional radius of the wire is 2.50 mm, and the wire makes an angle of 30.0° with the rod. (a) What is the length of the wire? (b) An object of mass 90.0 kg is suspended from the right-hand end of the rod. What is the increase in the length of the wire when this object is added? In your analysis do you need to be concerned that the lengthening of the wire means that the rod is no longer horizontal?

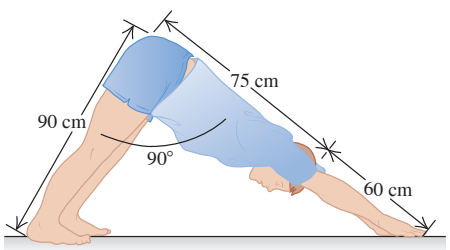
11.66 • A holiday decoration consists of two shiny glass spheres with masses 0.0240 kg and 0.0360 kg suspended from a uniform rod with mass 0.120 kg and length 1.00 m (**Fig. P11.66**). The rod is suspended from the ceiling by a vertical cord at each end, so that it is horizontal. Calculate the tension in each of the cords *A* through *F*.

Figure **P11.66**



11.67 •• BIO **Downward-Facing Dog.** The yoga exercise "Downward-Facing Dog" requires stretching your hands straight out above your head and bending down to lean against the floor. This exercise is performed by a 750 N person as shown in **Fig. P11.67**. When he bends his body at the hip to a 90° angle between his legs and trunk, his legs, trunk, head, and arms have the dimensions indicated. Furthermore, his legs and feet weigh a total of 277 N, and their center of mass is 41 cm from his hip, measured along his legs. The person's trunk, head, and arms weigh 473 N, and their center of gravity is 65 cm from his hip, measured along the upper body. (a) Find the normal force that the floor exerts on each foot and on each hand, assuming that the person does not favor either hand or either foot. (b) Find the friction force on each foot and on each hand, assuming that it is the same on both feet and on both hands (but not necessarily the same on the feet as on the hands). [*Hint:* First treat his entire body as a system; then isolate his legs (or his upper body).]

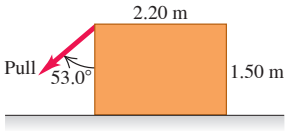
Figure **P11.67**



11.68 •• CP A brass wire is 1.40 m long and has a cross-sectional area of 6.00 mm^2. A small steel ball with mass 0.0800 kg is attached to the end of the wire. You hold the other end of the wire and whirl the ball in a vertical circle of radius 1.40 m. What speed must the ball have at the lowest point of its path if its fractional change in length of the brass wire at this point from its unstretched length is 2.0×10^{-5}? Treat the ball as a point mass.

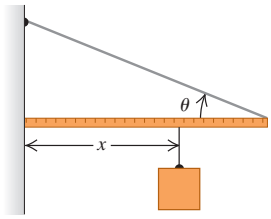
11.69 • A worker wants to turn over a uniform, 1250 N, rectangular crate by pulling at 53.0° on one of its vertical sides (**Fig. P11.69**). The floor is rough enough to prevent the crate from slipping. (a) What pull is needed to just start the crate to tip? (b) How hard does the floor push upward on the crate? (c) Find the friction force on the crate. (d) What is the minimum coefficient of static friction needed to prevent the crate from slipping on the floor?

Figure **P11.69**



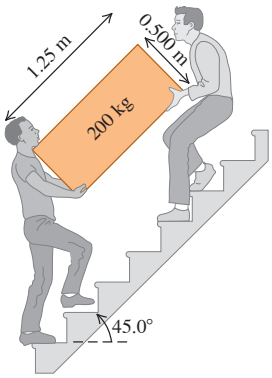
11.70 ••• One end of a uniform meter stick is placed against a vertical wall (**Fig. P11.70**). The other end is held by a lightweight cord that makes an angle θ with the stick. The coefficient of static friction between the end of the meter stick and the wall is 0.40. (a) What is the maximum value the angle θ can have if the stick is to remain in equilibrium? (b) Let the angle θ be 18°. A block of the same weight as the meter stick is suspended from the stick, as shown, at a distance x from the wall. What is the minimum value of x for which the stick will remain in equilibrium? (c) When $\theta = 18°$, how large must the coefficient of static friction be so that the block can be attached 10 cm from the left end of the stick without causing it to slip?

Figure **P11.70**



11.71 •• Two friends are carrying a 200 kg crate up a flight of stairs. The crate is 1.25 m long and 0.500 m high, and its center of gravity is at its center. The stairs make a 45.0° angle with respect to the floor. The crate also is carried at a 45.0° angle, so that its bottom side is parallel to the slope of the stairs (**Fig. P11.71**). If the force each person applies is vertical, what is the magnitude of each of these forces? Is it better to be the person above or below on the stairs?

Figure **P11.71**



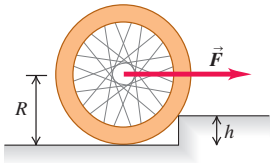
11.72 ••• In a city park a nonuniform wooden beam 4.00 m long is suspended horizontally by a light steel cable at each end. The cable at the left-hand end makes an angle of 30.0° with the vertical and has tension 620 N. The cable at the right-hand end of the beam makes an angle of 50.0° with the vertical. As an employee of the Parks and Recreation Department, you are asked to find the weight of the beam and the location of its center of gravity.

11.73 •• CALC BIO Refer to the discussion of holding a dumbbell in Example 11.4 (Section 11.3). The maximum weight that can be held in this way is limited by the maximum allowable tendon tension T (determined by the strength of the tendons) and by the distance D from the elbow to where the tendon attaches to the forearm. (a) Let T_{max} represent the maximum value of the tendon tension. Use the results of Example 11.4 to express w_{max} (the maximum weight that can be held) in terms of T_{max}, L, D, and h. Your expression should *not* include the angle θ. (b) The tendons of different primates are attached to the forearm at different values of D. Calculate the derivative of w_{max} with respect to D, and

determine whether the derivative is positive or negative. (c) A chimpanzee tendon is attached to the forearm at a point farther from the elbow than for humans. Use this to explain why chimpanzees have stronger arms than humans. (The disadvantage is that chimpanzees have less flexible arms than do humans.)

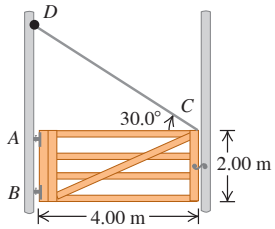
11.74 •• You are trying to raise a bicycle wheel of mass m and radius R up over a curb of height h. To do this, you apply a horizontal force $\vec{F}$ (**Fig. P11.74**). What is the smallest magnitude of the force $\vec{F}$ that will succeed in raising the wheel onto the curb when the force is applied (a) at the center of the wheel and (b) at the top of the wheel? (c) In which case is less force required?

Figure **P11.74**



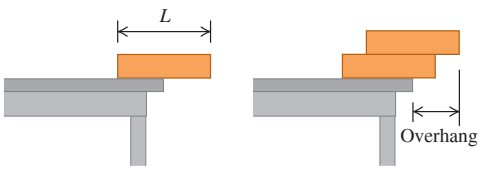
11.75 • **The Farmyard Gate.** A gate 4.00 m wide and 2.00 m high weighs 550 N. Its center of gravity is at its center, and it is hinged at A and B. To relieve the strain on the top hinge, a wire CD is connected as shown in **Fig. P11.75**. The tension in CD is increased until the horizontal force at hinge A is zero. What is (a) the tension in the wire CD; (b) the magnitude of the horizontal component of the force at hinge B; (c) the combined vertical force exerted by hinges A and B?

Figure **P11.75**



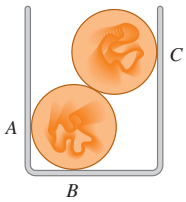
11.76 • If you put a uniform block at the edge of a table, the center of the block must be over the table for the block not to fall off. (a) If you stack two identical blocks at the table edge, the center of the top block must be over the bottom block, and the center of gravity of the two blocks together must be over the table. In terms of the length L of each block, what is the maximum overhang possible (**Fig. P11.76**)? (b) Repeat part (a) for three identical blocks and for four identical blocks. (c) Is it possible to make a stack of blocks such that the uppermost block is not directly over the table at all? How many blocks would it take to do this? (Try.)

Figure **P11.76**



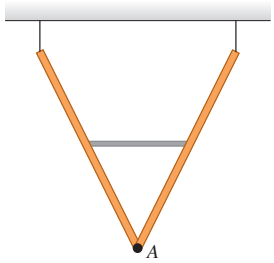
11.77 ••• Two uniform, 75.0 g marbles 2.00 cm in diameter are stacked as shown in **Fig. P11.77** in a container that is 3.00 cm wide. (a) Find the force that the container exerts on the marbles at the points of contact A, B, and C. (b) What force does each marble exert on the other?

Figure **P11.77**



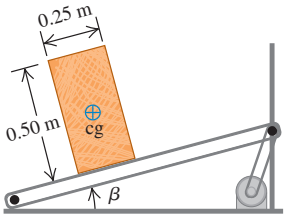
11.78 •• Two identical, uniform beams weighing 260 N each are connected at one end by a frictionless hinge. A light horizontal crossbar attached at the midpoints of the beams maintains an angle of 53.0° between the beams. The beams are suspended from the ceiling by vertical wires such that they form a "V" (**Fig. P11.78**). (a) What force does the crossbar exert on each beam? (b) Is the crossbar under tension or compression? (c) What force (magnitude and direction) does the hinge at point A exert on each beam?

Figure **P11.78**



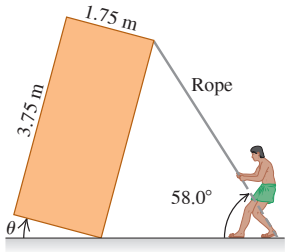
11.79 • An engineer is designing a conveyor system for loading hay bales into a wagon (**Fig. P11.79**). Each bale is 0.25 m wide, 0.50 m high, and 0.63 m long (the dimension perpendicular to the plane of the figure), with mass 30.8 kg. The center of gravity of each bale is at its geometrical center. The coefficient of static friction between a bale and the conveyor belt is 0.67, and the belt moves with constant speed. (a) The angle β of the conveyor is slowly increased. At some critical angle a bale will tip (if it doesn't slip first), and at some different critical angle it will slip (if it doesn't tip first). Find the two critical angles and determine which happens at the smaller angle. (b) Would the outcome of part (a) be different if the coefficient of friction were 0.40?

Figure **P11.79**



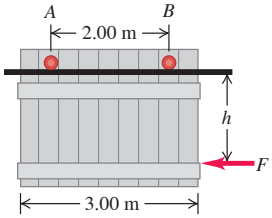
11.80 ••• **Pyramid Builders.** Ancient pyramid builders are balancing a uniform rectangular slab of stone tipped at an angle θ above the horizontal using a rope (**Fig. P11.80**). The rope is held by five workers who share the force equally. (a) If $\theta = 16.6°$, what force does each worker exert on the rope? (b) As θ increases, does each worker have to exert more or less force than in part (a), assuming they do not change the angle of the rope? Why? (c) At what angle do the workers need to exert *no force* to balance the slab? What happens if θ exceeds this value?

Figure **P11.80**



11.81 •• A garage door is mounted on an overhead rail (**Fig. P11.81**). The wheels at A and B have rusted so that they do not roll, but rather slide along the track. The coefficient of kinetic friction is 0.49. The distance between the wheels is 2.00 m, and each is 0.50 m from the vertical sides of the door. The door is uniform and weighs 958 N. It is pushed to the left at constant speed by a horizontal force $\vec{F}$. (a) If the distance h is 1.43 m, what is the vertical component of the force exerted on each wheel by the track? (b) Find the maximum value h can have without causing one wheel to leave the track.

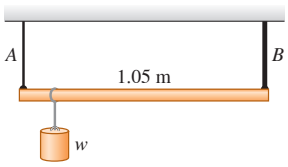
Figure **P11.81**



11.82 ••• CP A 11.1 kg mass, fastened to the end of an aluminum wire with an unstretched length of 0.54 m, is whirled in a vertical circle with a constant angular speed of 130 rev/min. The cross-sectional area of the wire is 1.3×10^{-2} cm^2. Calculate the elongation of the wire when the mass is (a) at the lowest point of the path and (b) at the highest point of its path.

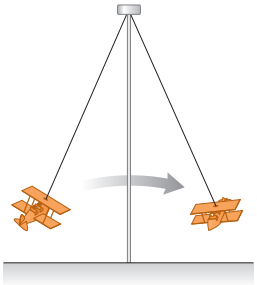
11.83 ••• A 1.05-m-long rod of negligible weight is supported at its ends by wires A and B of equal length (**Fig. P11.83**). The cross-sectional area of A is 1.96 mm^2 and that of B is 4.29 mm^2. Young's modulus for wire A is 2.50×10^{11} Pa; that for B is 1.10×10^{11} Pa. At what point along the rod should a weight w be suspended to produce (a) equal stresses in A and B and (b) equal strains in A and B?

Figure P11.83



11.84 ••• CP An amusement park ride consists of airplane-shaped cars attached to steel rods (**Fig. P11.84**). Each rod has a length of 16.0 m and a cross-sectional area of 8.20 cm^2. (a) How much is each rod stretched when it is vertical and the ride is at rest? (Assume that each car plus two people seated in it has a total weight of 1830 N.) (b) When operating, the ride has a maximum angular speed of 8.6 rev/min. How much is the rod stretched then?

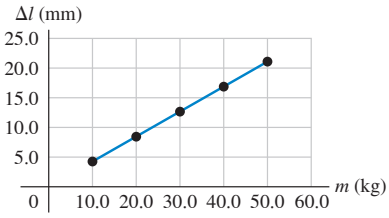
Figure P11.84



11.85 ••• CP BIO **Stress on the Shin Bone.** The compressive strength of our bones is important in everyday life. Young's modulus for bone is about 1.4×10^{10} Pa. Bone can take only about a 1.0% change in its length before fracturing. (a) What is the maximum force that can be applied to a bone whose minimum cross-sectional area is 3.0 cm^2? (This is approximately the cross-sectional area of a tibia, or shin bone, at its narrowest point.) (b) Estimate the maximum height from which a 70 kg man could jump and not fracture his tibia. Take the time between when he first touches the floor and when he has stopped to be 0.030 s, and assume that the stress on his two legs is distributed equally.

11.86 •• DATA You are to use a long, thin wire to build a pendulum in a science museum. The wire has an unstretched length of 22.0 m and a circular cross section of diameter 0.860 mm; it is made of an alloy that has a large breaking stress. One end of the wire will be attached to the ceiling, and a 9.50 kg metal sphere will be attached to the other end. As the pendulum swings back and forth, the wire's maximum angular displacement from the vertical will be 36.0°. You must determine the maximum amount the wire will stretch during this motion. So, before you attach the metal sphere, you suspend a test mass (mass m) from the wire's lower end. You then measure the increase in length Δl of the wire for several different test masses. **Figure P11.86**, a graph of Δl versus m, shows the results and the straight line that gives the best fit to the data. The equation for this line is $\Delta l = (0.422 \text{ mm/kg})m$. (a) Assume that $g = 9.80$ m/s^2, and use Fig. P11.86 to calculate Young's modulus Y for this wire. (b) You remove the test masses, attach the 9.50 kg sphere, and release the sphere from rest, with the wire displaced by 36.0°. Calculate the amount the wire will stretch as it swings through the vertical. Ignore air resistance.

Figure P11.86



11.87 •• DATA You need to measure the mass M of a 4.00-m-long bar. The bar has a square cross section but has some holes drilled along its length, so you suspect that its center of gravity isn't in the middle of the bar. The bar is too long for you to weigh on your scale. So, first you balance the bar on a knife-edge pivot and determine that the bar's center of gravity is 1.88 m from its left-hand end. You then place the bar on the pivot so that the point of support is 1.50 m from the left-hand end of the bar. Next you suspend a 2.00 kg mass (m_1) from the bar at a point 0.200 m from the left-hand end. Finally, you suspend a mass $m_2 = 1.00$ kg from the bar at a distance x from the left-hand end and adjust x so that the bar is balanced. You repeat this step for other values of m_2 and record each corresponding value of x. The table gives your results.

m_2 (kg)	1.00	1.50	2.00	2.50	3.00	4.00
x (m)	3.50	2.83	2.50	2.32	2.16	2.00

(a) Draw a free-body diagram for the bar when m_1 and m_2 are suspended from it. (b) Apply the static equilibrium equation $\sum \tau_z = 0$ with the axis at the location of the knife-edge pivot. Solve the equation for x as a function of m_2. (c) Plot x versus $1/m_2$. Use the slope of the best-fit straight line and the equation you derived in part (b) to calculate that bar's mass M. Use $g = 9.80$ m/s^2. (d) What is the y-intercept of the straight line that fits the data? Explain why it has this value.

11.88 ••• DATA You are a construction engineer working on the interior design of a retail store in a mall. A 2.00-m-long uniform bar of mass 8.50 kg is to be attached at one end to a wall, by means of a hinge that allows the bar to rotate freely with very little friction. The bar will be held in a horizontal position by a light cable from a point on the bar (a distance x from the hinge) to a point on the wall above the hinge. The cable makes an angle θ with the bar. The architect has proposed four possible ways to connect the cable and asked you to assess them:

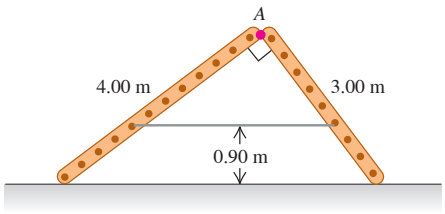
Alternative	A	B	C	D
x (m)	2.00	1.50	0.75	0.50
θ (degrees)	30	60	37	75

(a) There is concern about the strength of the cable that will be required. Which set of x and θ values in the table produces the smallest tension in the cable? The greatest? (b) There is concern about the breaking strength of the sheetrock wall where the hinge will be attached. Which set of x and θ values produces the smallest horizontal component of the force the bar exerts on the hinge? The largest? (c) There is also concern about the required strength of the hinge and the strength of its attachment to the wall. Which set of x and θ values produces the smallest magnitude of the vertical component of the force the bar exerts on the hinge? The largest? (*Hint:* Does the direction of the vertical component of the force the hinge exerts on the bar depend on where along the bar the cable is attached?) (d) Is one of the alternatives given in the table preferable? Should any of the alternatives be avoided? Discuss.

CHALLENGE PROBLEMS

11.89 ••• Two ladders, 4.00 m and 3.00 m long, are hinged at point A and tied together by a horizontal rope 0.90 m above the floor (**Fig. P11.89**). The ladders weigh 480 N and 360 N, respectively, and the center of gravity of each is at its center. Assume that the floor is freshly waxed and frictionless. (a) Find the upward force at the bottom of each ladder. (b) Find the tension in the rope. (c) Find the magnitude

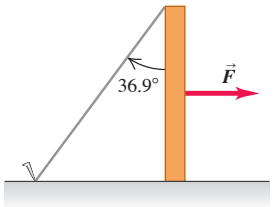
Figure P11.89



of the force one ladder exerts on the other at point A. (d) If an 800 N painter stands at point A, find the tension in the horizontal rope.

11.90 ••• **Knocking Over a Post.** One end of a post weighing 400 N and with height h rests on a rough horizontal surface with $\mu_s = 0.30$. The upper end is held by a rope fastened to the surface and making an angle of 36.9° with the post (**Fig. P11.90**). A horizontal force $\vec{F}$ is exerted on the post as shown. (a) If the force $\vec{F}$ is applied at the midpoint of the post, what is the largest value it can have without causing the post to slip? (b) How large can the force be without causing the post to slip if its point of application is $\frac{6}{10}$ of the way from the ground to the top of the post? (c) Show that if the point of application of the force is too high, the post cannot be made to slip, no matter how great the force. Find the critical height for the point of application.

Figure P11.90

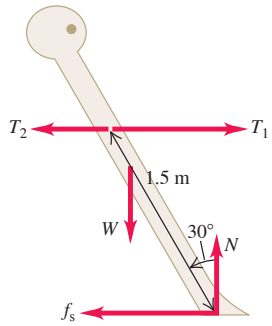


11.91 ••• CP An angler hangs a 4.50 kg fish from a vertical steel wire 1.50 m long and 5.00×10^{-3} cm^2 in cross-sectional area. The upper end of the wire is securely fastened to a support. (a) Calculate the amount the wire is stretched by the hanging fish. The angler now applies a varying force $\vec{F}$ at the lower end of the wire, pulling it very slowly downward by 0.500 mm from its equilibrium position. For this downward motion, calculate (b) the work done by gravity; (c) the work done by the force $\vec{F}$; (d) the work done by the force the wire exerts on the fish; and (e) the change in the elastic potential energy (the potential energy associated with the tensile stress in the wire). Compare the answers in parts (d) and (e).

MCAT-STYLE PASSAGE PROBLEMS

BIO Torques and Tug-of-War. In a study of the biomechanics of the tug-of-war, a 2.0-m-tall, 80.0 kg competitor in the middle of the line is considered to be a rigid body leaning back at an angle of 30.0° to the vertical. The competitor is pulling on a rope that is held horizontal a distance of 1.5 m from his feet (as measured along the line of the body). At the moment shown in the figure, the man is stationary and the tension in the rope in front of him is $T_1 = 1160$ N. Since there is friction between the rope and his hands, the tension in the rope behind him, T_2, is not equal to T_1. His center of mass is halfway between his feet and the top of his head. The coefficient of static friction between his feet and the ground is 0.65.

Figure 11.92 Side view.



11.92 What is tension T_2 in the rope behind him? (a) 590 N; (b) 650 N; (c) 860 N; (d) 1100 N.

11.93 If he leans slightly farther back (increasing the angle between his body and the vertical) but remains stationary in this new position, which of the following statements is true? Assume that the rope remains horizontal. (a) The difference between T_1 and T_2 will increase, balancing the increased torque about his feet that his weight produces when he leans farther back; (b) the difference between T_1 and T_2 will decrease, balancing the increased torque about his feet that his weight produces when he leans farther back; (c) neither T_1 nor T_2 will change, because no other forces are changing; (d) both T_1 and T_2 will change, but the difference between them will remain the same.

11.94 His body is again leaning back at 30.0° to the vertical, but now the height at which the rope is held above—but still parallel to—the ground is varied. The tension in the rope in front of the competitor (T_1) is measured as a function of the shortest distance between the rope and the ground (the holding height). Tension T_1 is found to decrease as the holding height increases. What could explain this observation? As the holding height increases, (a) the moment arm of the rope about his feet decreases due to the angle that his body makes with the vertical; (b) the moment arm of the weight about his feet decreases due to the angle that his body makes with the vertical; (c) a smaller tension in the rope is needed to produce a torque sufficient to balance the torque of the weight about his feet; (d) his center of mass moves down to compensate, so less tension in the rope is required to maintain equilibrium.

11.95 His body is leaning back at 30.0° to the vertical, but the coefficient of static friction between his feet and the ground is suddenly reduced to 0.50. What will happen? (a) His entire body will accelerate forward; (b) his feet will slip forward; (c) his feet will slip backward; (d) his feet will not slip.

ANSWERS

Chapter Opening Question ?

(i) Each stone in the arch is under compression, not tension. This is because the forces on the stones tend to push them inward toward the center of the arch and thus squeeze them together. Compared to a solid supporting wall, a wall with arches is just as strong yet much more economical to build.

Key Example VARIATION Problems

VP11.1.1 13.3 kg
VP11.1.2 11.0 m
VP11.1.3 $L/10$ to the right of center
VP11.1.4 141 kg
VP11.4.1 5.86 m
VP11.4.2 nose wheel: 19.8%, main wheels: 80.2%
VP11.4.3 (a) $F_{\text{hinge-}y} = 0$ (b) $T = w/\sin\theta$ (c) $F_{\text{hinge-}x} = w/\tan\theta$
VP11.4.4 (a) $F_{\text{hinge-}y} = w/2$ (b) $T = 3w/2\sin\theta$ (c) $F_{\text{hinge-}x} = 3w/2\tan\theta$
VP11.7.1 (a) 5.5×10^8 Pa (b) 3.5×10^4 N
VP11.7.2 (a) 2.3×10^{-4} (b) 0.19 mm
VP11.7.3 (a) 4.4×10^{-9} m^3 (b) 6.7×10^{-9} m^3
VP11.7.4 4.8×10^{-5} m

Bridging Problem

(a) $T = \dfrac{2mg}{3\sin\theta}$

(b) $F = \dfrac{2mg}{3\sin\theta}\sqrt{\cos^2\theta + \frac{1}{4}\sin^2\theta}, \quad \phi = \arctan\left(\frac{1}{2}\tan\theta\right)$

(c) $\Delta l = \dfrac{mgl_0}{2AY\tan\theta}$

(d) 4